🕸️ GNN-AML Detection: Relational Intelligence & Network Topology

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Target Stakeholders: AML Compliance Units, Financial Crime (FinCrime) Operations, & Chief Risk Officers (CRO)

📊 Executive Summary

The **GNN-AML Detection Suite** is a high-fidelity Proof of Concept (PoC) designed to solve the "Structuring Gap." While legacy systems rely on static $10,000 reporting thresholds—which are easily bypassed by breaking funds into smaller, connected transfers—this platform ues **Graph Neural Networks (GNNs)** to identify illicit behavior based on network topology and transaction velocity.

Strategic Business Value:

- **Beyond the $10k Threshold:** Detects "smurfing" networks where multiple actors coordinate small deposits to evade traditional triggers.
- **False Positive Reduction:** AI-driven RegTech is projected to cut false positives by up to 40% by 2026, significantly reducing the operational burden on compliance investigators.
- **Regulatory Resilience:** Aligns with the **2026 AML Compliance Revolution**, moving from reactive responses to "proactive early warning" systems for Basel IV and FATF standards.

🚀 Strategic Modules & Intelligence

The portal features a "Human-in-the-Loop" architecture, providing investigators with a multi-dimensional view of relational risk.

🧠 The Relational Brain (GraphSAGE Engine)

- **Goal:** Learn node representations from large-scale, evolving transaction graphs.
- **Logic:** Uses a **GraphSAGE** (Sample and Aggregate) architecture to aggregate feature information from a node's local neighborhood, allowing for high-precision detection even in "cold start" scenarios.
- **Training:** Trained on 203,000+ real-world transactions from the **Elliptic Dataset**, reaching precision benchmarks up to 98%.

🕸️ Interactive Network Topology (The "Eyes")

- **Goal:** Visualize the hidden links between seemingly unrelated accounts.
- **Analytics:** A Streamlit-based interface that maps "smurfing" chains, identifying illicit clusters and the flow of funds through layering and integration phases.

⚖️ Automated Compliance Triage

- **Goal:** Rapidly convert high-risk signals into actionable Suspicious Activity Reports (SARs).
- **Explainability:** Incorporates feature importance plots and node-level risk scoring to show auditors exactly *why* a specific network cluster was flagged.

🛠️ Technical Architecture

This project demonstrates proficiency in deep learning on graph-structured data and high-performance financial modeling.

Layer	Technology Stack
Frontend	Streamlit (Session state management & network ledger visualization)
GNN Engine	`PyTorch Geometric` (GraphSAGE), `Scikit-Learn` (Evaluation metrics)
Data Logic	`Pandas`, `NumPy` (166-feature transaction vectorization)
Environment	GPU-Accelerated (NVIDIA Tesla) Cloud Infrastructure for high-scale inference.

📊 Strategic KPI Glossary

Metrics calculated within the suite are aligned with **2026 Global AML Trends**:

- **SAR Conversion Rate:** The percentage of AI alerts that result in a filed Suspicious Activity Report (SAR).
- **Smurfing Detection Lift:** The delta in illicit pattern identification compared to legacy rule-based systems.
- **Precision (Illicit Class):** Targeted at 85%+ to ensure high-confidence flags for the legal and compliance teams.

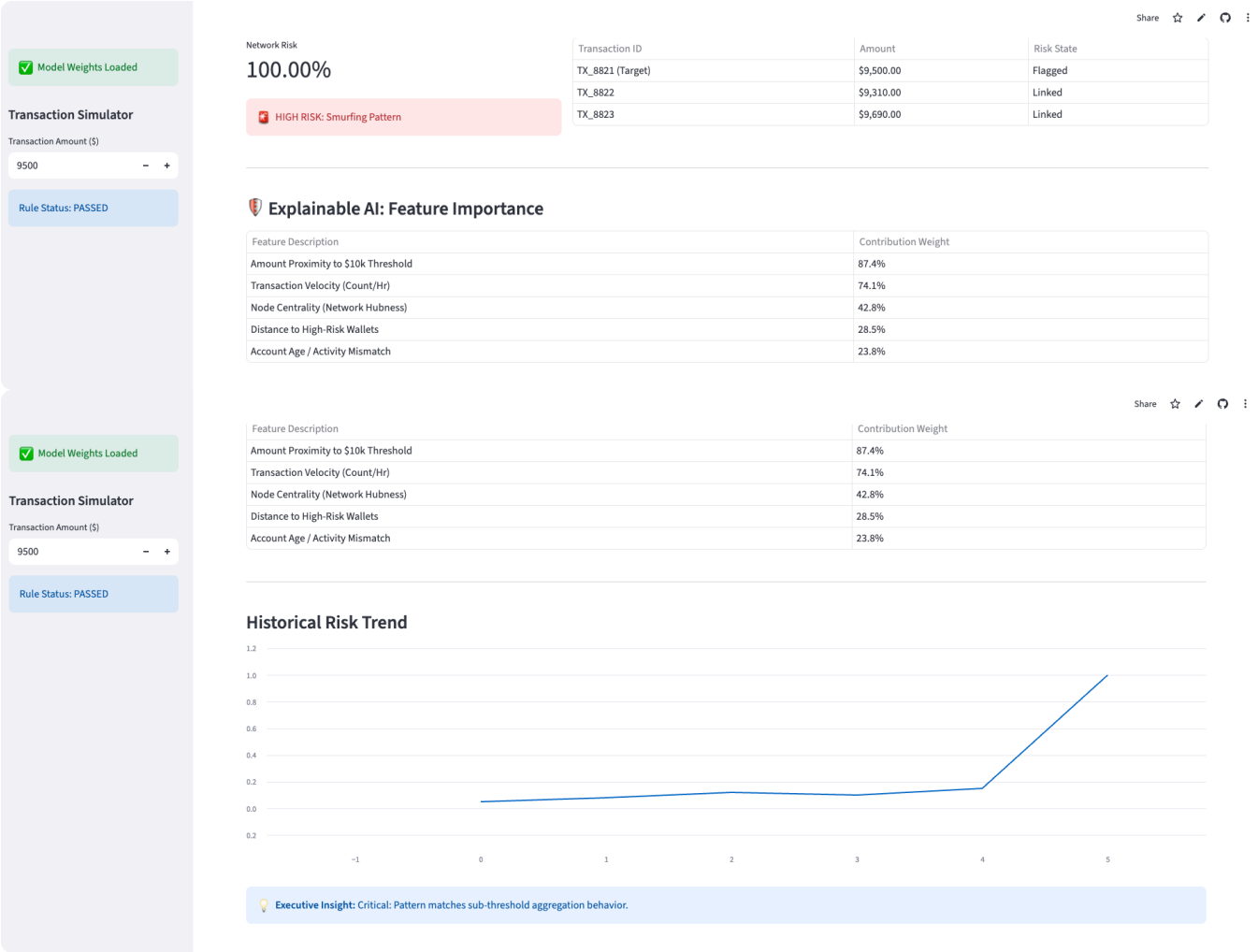
💻 Installation & Usage

To launch the GNN-AML Portal locally:

1. **Clone the repository:** `git clone https://github.com/your-username/gnn-aml-detection-poc.git`
2. **Install dependencies:** `pip install -r requirements.txt`
3. **Launch the application:** `streamlit run app.py`

📁 Repository Strategy

This repository serves as a **Narrative of Authority**, demonstrating that strategic AML defenses must be grounded in relational topology rather than static data.

Operational Focus (Old)	Strategic Focus (New)
Focuses on $10k dollar thresholds.	Focuses on Network Topology & Node Embedding.
Technical stack is a simple list.	Architectural narrative of "Scalable Relational Intelligence."
Feature list of "Dashboard."	Strategic "Relational Brain" for CRO oversight.

Screenshots